Q1.

This question is about elements and the periodic table.

(a) Use the correct answers from the box to complete the sentences.

atoms	atomic weights	electrons	proton numbers

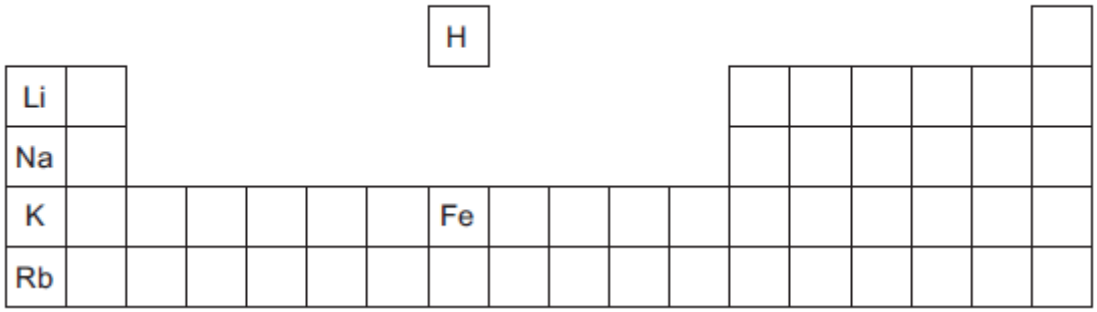
Newlands' and Mendeleev's periodic tables show the elements in order of their _____ .

Following the discovery of protons and _____, the modern periodic table shows the elements in order of their _____ .

(3)

(b) **Figure 1** shows the position of six elements in the modern periodic table.

Figure 1



(i) Which **one** of these six elements has the lowest boiling point?

_____

(1)

(ii) Complete the sentence.

In the periodic table, rubidium (Rb) is in Group _____ .

(1)

(iii) Which of these three elements is the most reactive?

Tick (✓) **one** box.

Lithium (Li) ☐

Sodium (Na) ☐

Potassium (K) ☐

(1)

(iv) Which **two** statements are correct?

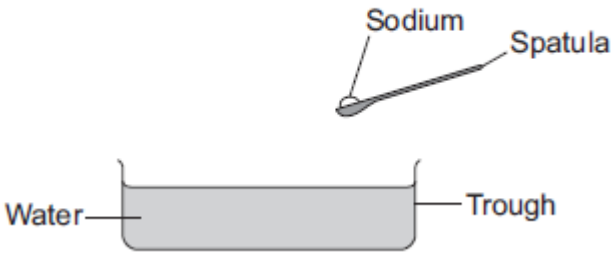
Tick (✓) **two** boxes.

Iron has a higher density than potassium.	☐
Iron is softer than potassium.	☐
Iron reacts vigorously with water.	☐
Iron forms ions that have different charges.	☐

(2)

(c) **Figure 2** shows sodium being put into water.

Figure 2



Describe **three** observations that can be seen when sodium is put into water.

1. _____

_____

2. _____

_____

3. _____

_____

(3)

(Total 11 marks)

Q1.

(a) atomic weights

must be in this order

1

electrons

1

proton numbers

1

(b) (i) H/hydrogen

allow H_2 or h

1

(ii) one / 1

allow alkali metals

1

(iii) Potassium (K)

1

(iv) Iron has a higher density than potassium

1

Iron forms ions that have different charges

1

(c) any **three** from:

- melts
- fizzes / bubbles / effervesces

  allow gas produced
- sodium floats
- size of the sodium decreases

  allow dissolves / disappears
- sodium moves

  allow two marks for moves around on the surface of the water

3

[11]